

FACTORS INFLUENCING GI ABSORPTION OF DRUGS



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FACTORS INFLUENCING GI ABSORPTION OF DRUGS

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- The oral absorption process of drug from a pharmaceutical DF is very complex.
- However, the major steps occurring during oral drug absorption can be regarded as part of a serial process:
 - The dissolution of the drug from the DF
 - The solubility of drug as a function of its physicochemical characteristics
 - The drug's effective permeability to the intestinal mucosa and
 - The drug's pre-systemic metabolism.

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- ❑ There are many factors that may affect the above processes, and finally affect the **rate** and **extent** of oral drug absorption.
- ❑ These factors can be divided into three categories:-
 - Physicochemical properties of a drug,
 - ✓ solubility
 - ✓ particle size
 - ✓ polymorphism,
 - ✓ pKa,
 - ✓ lipophilicity, and so on.

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- Formulation factors,
 - solution, capsule, tablet, suspension and so on.
 - nature of excipients
 - manufacturing procedures

- Physiological factors,
 - gastrointestinal pH,
 - gastric emptying,
 - small intestinal transit time,
 - blood flow

PHYSICOCHEMICAL FACTORS INFLUENCING DRUG ABSORPTION

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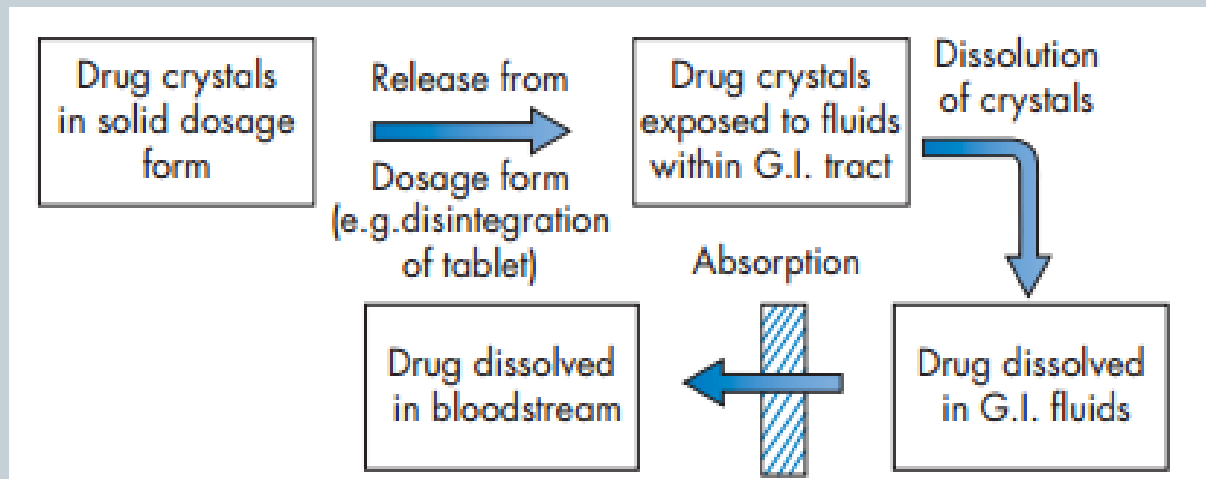
- Extensive experimentation has shown that the primary physicochemical properties of a drug influencing its passive absorption into and across biological membranes are:
 - oil/water partition coefficient,
 - Extent of ionization in biological fluids determined by its pKa and pH of the fluid,
 - molecular weight.

PHYSICOCHEMICAL FACTORS...

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Dissolution and Solubility

- In order for absorption to occur, a drug or a therapeutic agent must be present in solution form.
- This means that drugs administered orally in solid DFs must dissolve in the GI fluids before absorption can occur



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- Precondition for absorption of solid drugs
 - **Dissolution**: rate of solution of solid in solvent
 - **Solubility**: extent to which dissolution proceeds under given set of conditions
- A drug is defined as being poorly soluble when its dissolution rate is so slow that dissolution takes longer than the transit time past its absorptive sites, resulting in incomplete oral absorption.

PHYSICOCHEMICAL FACTORS...

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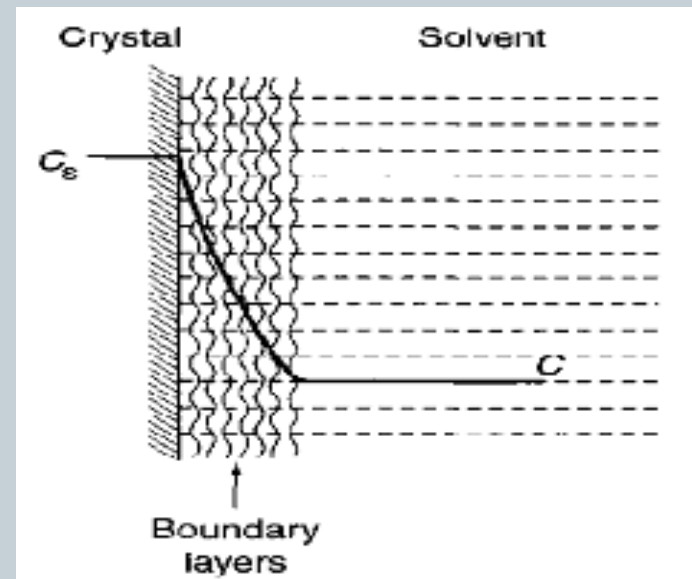
- Dissolution of solid in liquid is composed of two stages:
 - I. Interfacial reaction liberates solute molecules from solid phase
 - Involves phase change (molecules of solid become molecules of solute in solvent)
 - Solution in contact with solid will be saturated (concentration will be C_s , saturated solution)

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II. Solute molecules migrate through boundary layers to bulk of solution

- Once solute passes boundary layer, rapid mixing occurs and conc. gradient is destroyed and concentration will be C
- Conc. of solution in boundary layer changes from being C_s at crystal surface to bulk conc, C at its outer most limit.



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- Rate of diffusion of dissolved solute across boundary layer determines rate of dissolution
 - Obey Fick's law of diffusion
 - Rate $\propto$ conc. difference b/n two sides of diffusion layer

$$dC/dt = K\Delta C$$

where

- k is dissolution rate constant
- ΔC is difference in conc. at solid surface (C_s) and bulk (C)
- Dissolution rate of a drug under sink conditions is directly proportional to its *intrinsic solubility* in the diffusion layer surrounding each dissolving drug particle, C_s

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Noyes-Whitney equation:

where

- D =diffusion coefficient of drug
- A =effective SA of drug particle
- h =thickness of diffusion layer

$$dC/dt = \frac{DA(C_s - C)}{h}$$

❖ The ***Noyes-Whitney*** equation tells us that the dissolution rate (***dC/dt***) of a drug in the GI tract depends on:

- ***diffusion coefficient (D) of a drug***
- ***surface area (A) of the undissolved solid drug***
- ***saturation solubility (Cs) of the drug in the GI fluid***
- ***thickness of the diffusion layer (h)***

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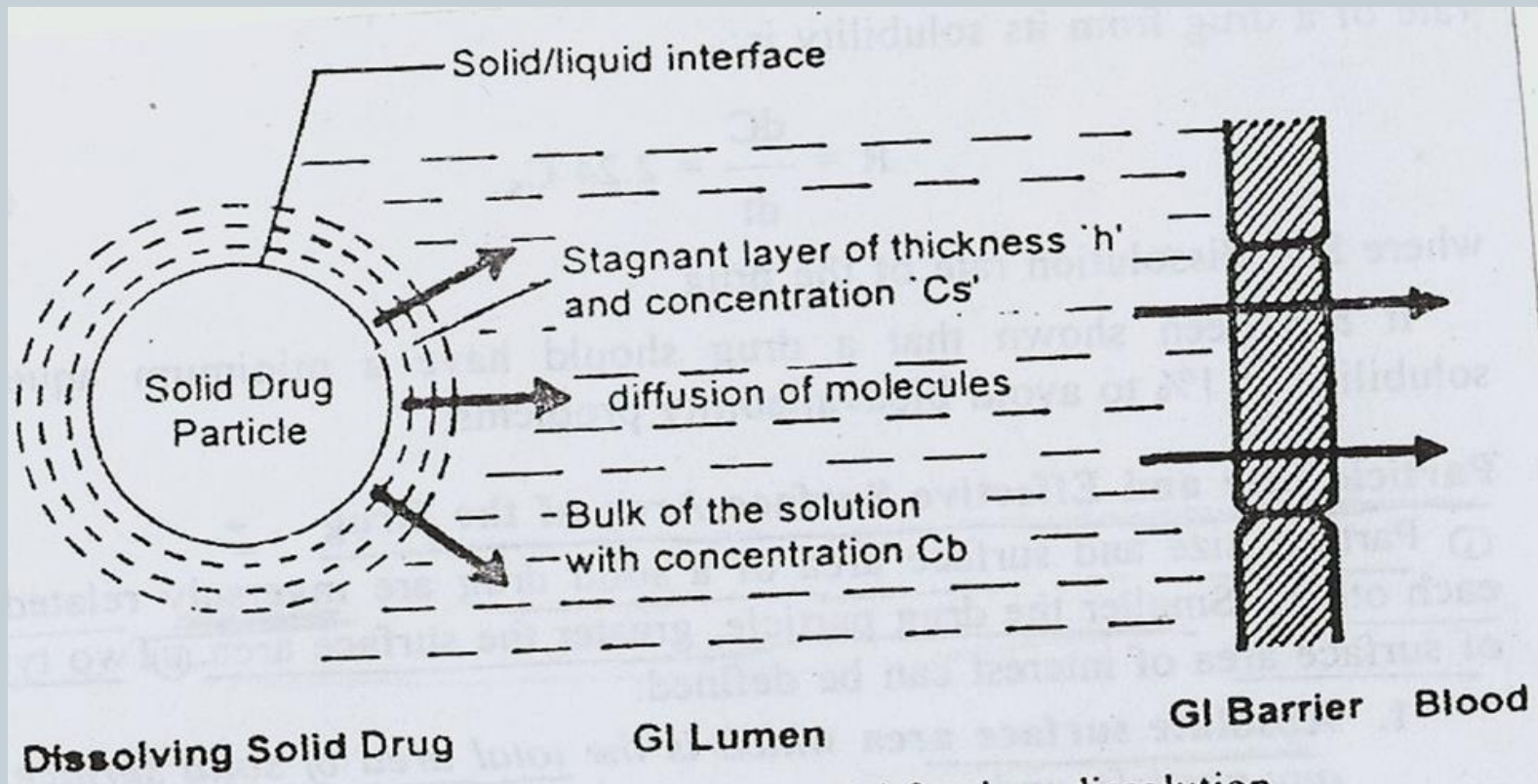


Figure: diffusion layer model for drug dissolution

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- Some of the more important factors that affect the dissolution rate, especially, of slowly dissolving or poorly soluble substances, are:
 - ✦ particle size and surface area
 - ✦ solubility of drug in the diffusion layer
 - ✦ the crystal form of a drug etc.
- Generally, for hydrophobic drugs the rate of absorption and **bioavailability** may be improved by increasing the rate of dissolution

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particle size and surface area

- A drug dissolves more rapidly when its surface area is increased.
- This increase in surface area is accomplished by reducing the particle size of the drug.
- *The smaller the particle size the larger the surface area and the faster the dissolution.*

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- Many poorly soluble, slowly dissolving drugs are presented in micronized form to increase SA:
 - Digoxin (Cardiac glycoside)
 - Medroxyprogesterone acetate (Hormone)
 - Ibuprofen (NSAID)
 - Griseofulvin
- **Griseofulvin:**
 - ❖ Reduction of particle size from ~ **10 μm** (specific SA= $0.4 \text{ m}^2/\text{g}$) to **2.7 μm** (specific SA= $1.5 \text{ m}^2/\text{g}$) produced approx. **doubled** amount of drug absorbed in humans.

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Polymorphism

- ❑ Depending upon the internal structure, a solid can exist either in a *crystalline* or *amorphous* form
- ❑ When a substance exists in more than one crystalline form, the different forms are designated as *polymorphs* and the phenomenon as *polymorphism*.
- ❑ *The amorphous form dissolves more rapidly than the corresponding crystalline form.*
 - ✓ amorphous form of *novobiocin* antibiotic was readily absorbed following oral administration of an aqueous suspension to humans.

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❑ Though chemically the same, polymorphs differ substantially with regards to physical properties.

✓ Solubility and dissolution rate will likely influence the rate of absorption

❑ Many drugs exist in more than one crystalline form

Eg., Chloramphenicol palmitate, cortisone acetate, tetracyclines

Chloramphenicol palmitate

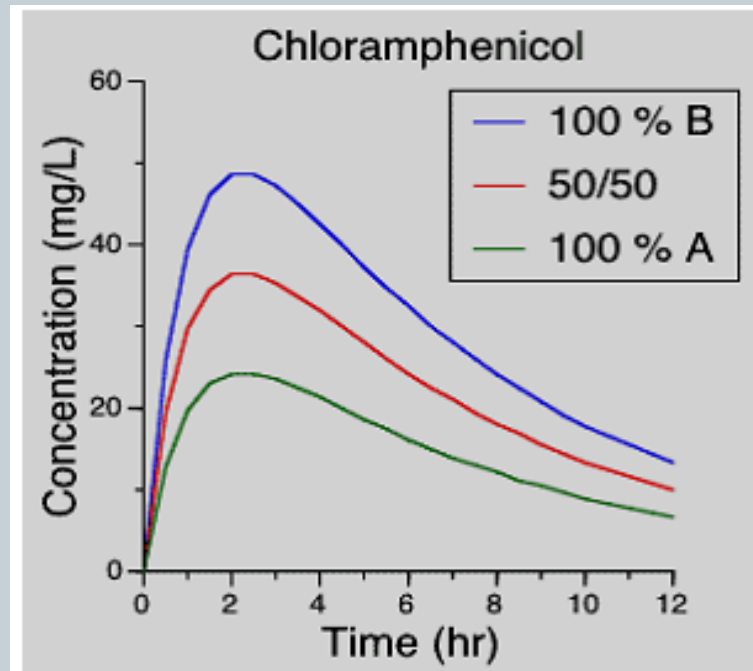
•Exists in 3 crystalline forms designated A, B and C

A	→	is the stable polymorph
B	→	is the metastable polymorph
C	→	is the unstable polymorph

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- Bioavailability studies of oral suspensions containing varying proportions of polymorphs A and B investigated
 - ✓ Extent of absorption increase in proportion of polymorph B
 - ✓ Rapid *in vivo* dissolution rate of polymorph B



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Salt form of the drug

- ❑ Most drugs are either *weak acids* or *weak bases*
- ❑ One of the easiest approach to enhance the solubility and dissolution rate of such drugs is to convert them into their salt forms.
- ❑ Salts of weak acids and weak bases generally have much higher aqueous solubility than the free acid or base.
- ❑ The dissolution rate of a weakly acidic drug in gastric fluid will be relatively low.

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- ❑ The drug salt form usually alters the drug dissolution rate by modifying **the diffusion layer pH** at the surface of the dissolving solid.
- ❑ If the pH in the diffusion layer **increased**, the solubility, C_s , of the acidic drug in this layer, and hence its dissolution rate in gastric fluids would be **increased**.
- ❑ The drug particles will dissolve at a **faster rate** and diffuse out of the diffusion layer into the bulk of the gastric fluid, where a lower bulk pH

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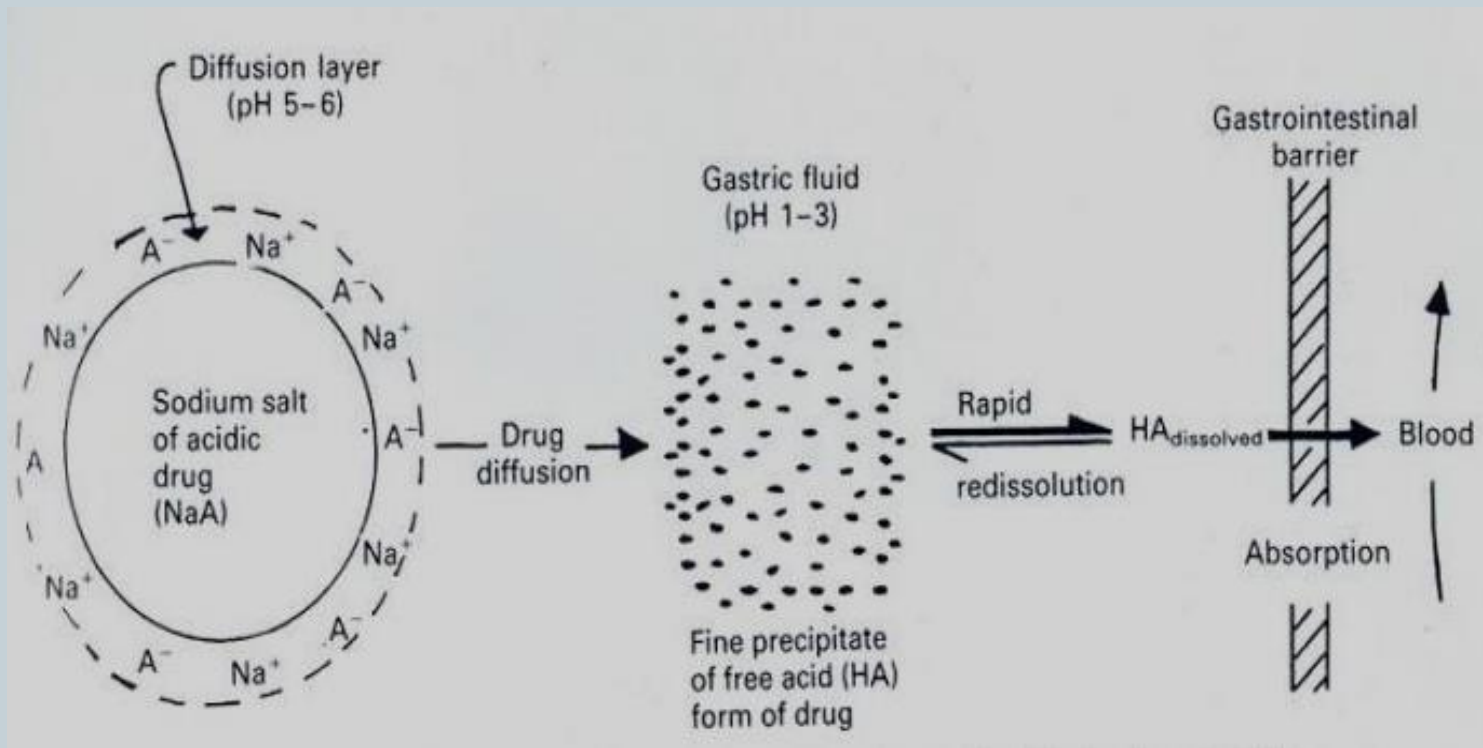


Figure: Dissolution process of a salt form of a weakly acidic drug in gastric fluid.

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pH - Partition Theory:

- The magnitude of the absorption of a drug following its availability as a solution is dictated by:
 - *dissociation constant*, expressed as pK_a ,
 - *lipid solubility* of a drug,
 - as well as the *pH* at the absorption site
- The interrelationship among these parameters (pH , pK_a and lipid solubility) is known as the *pH-partition theory* of drug absorption.
- The absorption of a **weak electrolyte** will be determined by the extent to which the drug exists in its *unionized form* at the site of absorption.

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- ❑ The extent to which a weakly acidic or basic drug ionizes in solution in the GI fluid may be calculated using *Henderson-Hasselbach equation*.
- ❑ *For a weakly acidic drug:*



$$\log \frac{A^-}{HA} = pH - pK_a$$

where:

-[HA] and [A⁻] are conc. of unionized and ionized forms

E.g. aspirin

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❑ *For a weakly basic drug:*



$$\log \frac{\text{BH}^+}{\text{B}} = \text{p}K_a - \text{pH}$$

where:

-[BH⁺] and [B] are conc. of ionized and unionized forms
E.g. chlorpromazine

Limitations of the pH-partition hypothesis:

- Despite their high degree of ionization, weak acids are highly absorbed from the small intestine and this may be due to:
 - 1- The large surface area
 - 2- A longer SI residence time.
 - 3- A microclimate pH, that exists on the surface of intestinal mucosa and is lower than that of the luminal pH of the SI.

PHYSICOCHEMICAL FACTORS...

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Oil/Water partition coefficient

- ❑ Number of drugs are poorly absorbed from GIT despite their unionized forms predominate
 - Lipid solubility
- ❑ The best parameter to correlate between water and lipid solubility is partition coefficient.
- ❑ Barbiturates: Barbitone and Thiopentone
 - Similar dissociation constants (pKa) 7.8 and 7.6
 - Similar degrees of ionization at intestinal pH
- ❖ Thiopentone is absorbed much better than barbitone!
 - More lipid soluble
 - Exhibit greater affinity for GI membrane

FORMULATION FACTORS INFLUENCING DRUG ABSORPTION

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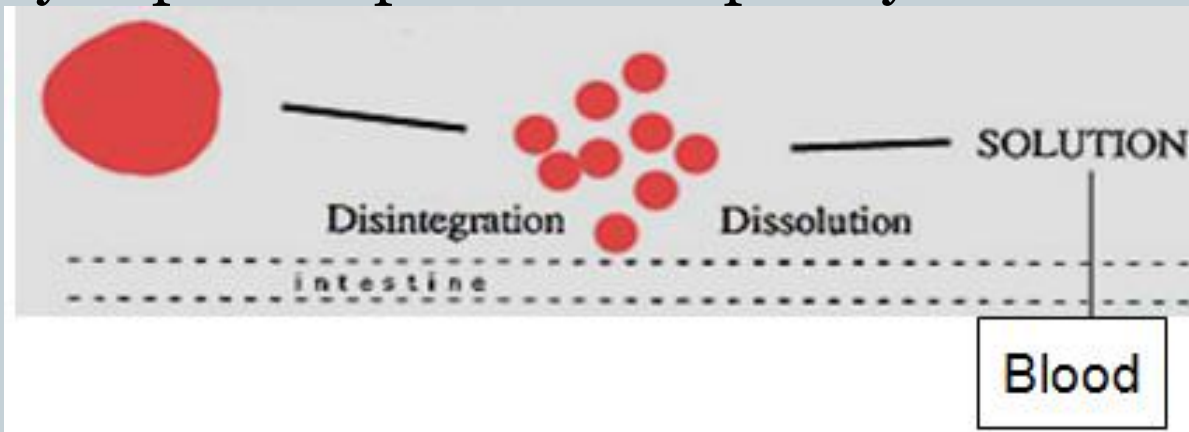
- ❑ Absorption rate depends also on:
 - the dosage form which is administered,
 - nature of ingredients/excipients used,
 - procedures used in formulation of dosage forms.

FORMULATION FACTORS...

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Nature and type of DF

- ❑ The relative rate at which a drug from a DF is presented to the body depends upon the complexity of DF.



- ❑ The availability of the drug for absorption from the dosage forms is in order.

solution > suspension > capsule > tablet > coated tablet.

FORMULATION FACTORS...

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Nature of ingredients/excipients:

- ❑ **Excipients** are added to a formulation to provide certain functional properties to the drug and dosage form.
- ❑ Some of these functional properties of the excipients are:
 - ❑ used to improve the compressibility of the active drug,
 - ❑ stabilize the drug against degradation,
 - ❑ decrease gastric irritation,
 - ❑ control the rate of drug absorption from the absorption site, etc.
- ❑ Excipients may be added intentionally to the formulation to enhance the rate and extent of drug absorption or to delay or slow the rate of drug absorption.

FORMULATION FACTORS...

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- ❑ Commonly used excipients in various DF include:
 - Vehicles, Diluents, Binders, Disintegrants, Lubricants, Suspending agents, Coatings, Colorants, Sweetners etc..
- ❑ Examples:
 - ✓ Tetracycline and DCP

FORMULATION FACTORS...

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Manufacturing variables

- ❑ Several manufacturing process influence drug dissolution from solid DFs.
- ❑ Process of such importance in the manufacturing of tablets and capsules are:
 - ✓ Method of granulation
 - ✓ Compression force
 - ✓ Intensity of packing of capsule contents

FORMULATION FACTORS...

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Disintegration time

- ❑ Is of particular importance in case of solid dosage forms like tablets and capsules
- ❑ Rapid disintegration-important in the therapeutic success of solid dosage form
- ❑ DT is directly related to the amount of binder present and the compression force of a tablet

PHYSIOLOGICAL FACTORS INFLUENCING DRUG ABSORPTION

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- ❑ The successful functioning of oral medication depends primarily on how the GIT processes drugs.
- ❑ Many factors are involved in oral drug delivery and the measured oral BA of a particular drug can be broken into components:
 - ❑ that reflect delivery to the intestine (gastric emptying, pH, food),
 - ❑ intestinal metabolism
 - ❑ first-pass hepatic extraction.
- ❑ All these factors play an important role in the performance of orally administered dosage forms, and can greatly contribute to the drug discovery process.

PHYSIOLOGICAL FACTORS...

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GI pH

- ❑ One important property of GI fluids is pH, which varies considerably along the length of the tract.
- ❑ The gastric fluid is highly acidic, usually ranging from pH 1 to 3.5 in the fasted state.
- ❑ The GI pH generally increases gradually as one moves down the stomach to the colon
- ❑ There is considerable inter subject variation in GI fluid pH, depending on:
 - ✓ the general health of the subject,
 - ✓ types of food ingested, and
 - ✓ drug therapy.

PHYSIOLOGICAL FACTORS...

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- ❑ GI fluid pH influence drug absorption in several ways:

A. pH dependent dissolution and absorption

- most drugs are weak acids or bases
 - the aqueous solubility of such compounds is influenced by pH and pK_a
 - the rate of dissolution from a DF, particularly tablets and capsules, is dependent on pH.
- ❑ Acidic drugs and basic drugs dissolve most readily in alkaline media and acidic solutions, respectively

PHYSIOLOGICAL FACTORS...

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- ❑ Since dissolution is a prerequisite step to absorption and is often the slowest process, especially for poorly water-soluble drugs, pH will exert a major influence on the overall absorption process.
- ❑ Since the major site of drug absorption is the small intestine, it would seem that poorly soluble basic drugs (e.g., ketoconazole and diazepam) must first dissolve in the acidic gastric fluids in order to be well absorbed from the intestine, as the dissolution rate in intestinal fluids will be low.

PHYSIOLOGICAL FACTORS...

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B. pH-dependent hydrolysis

❑ Numerous drugs are unstable in acidic media and will therefore degrade and provide lower effective doses depending on the pH of the gastric fluid, solubility of the drug, and the residence time of the DF in the stomach.

- Penicillin G

- Degradation depends on gastric residence time and pH

- ⇒ Gastric instability preclude oral use

- Erythromycin degrade rapidly at acidic pH

- Formulated as enteric-coated DFs!

PHYSIOLOGICAL FACTORS...

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Gastric emptying

- ❑ Most drugs are best absorbed from the small intestine, any factor that delays movement of drug from the stomach to small intestine will influence the rate (and possibly extent) of absorption
 - Gastric emptying may represent a limiting factor in drug absorption.
- ❑ Gastric emptying is quantitated by one of several measurements including
 - emptying time,
 - emptying half-life ($t_{50\%}$), and
 - emptying rate.

PHYSIOLOGICAL FACTORS...

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- ❖ *Emptying time* is the time needed for the stomach to empty the total initial stomach contents.
- ❖ *Emptying half-time* is the time it takes for the stomach to empty one half of its initial contents.
- ❖ *Emptying rate* is a measure of the speed of emptying.
- ❖ Factors that affect gastric emptying need to be understood because of the implications on drug absorption as well as on optimal DF design.
- ❖ Gastric emptying principally depends on:
 - DF type (liquid vs. solid)
 - fed/fasted state of stomach
 - postural position
 - co-administered drugs
 - disease state

PHYSIOLOGICAL FACTORS...

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❑ Factors delaying GE

- Fats in diet
- High viscosity of diet
- Lying on left side
- Disease: depression, hypothyroidism, gastric ulcer
- Drug: propantheline, atropine

❑ Factors promoting GE

- Fasting
- Lying on right side
- Disease: anxiety, hyperthyroidism
- Drug: Metoclopramide

PHYSIOLOGICAL FACTORS...

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Small Intestinal Transit

- ❑ Once a dosage form empties from the stomach and enter the small intestine, it will be exposed to an environment totally different from that of the stomach.
- ❑ The small intestine is the primary site of drug absorption,
 - the longer the residence time in this region, the greater the potential for complete absorption
- ❑ There are primarily two types of intestinal movements: ***propulsive*** and ***mixing***.

PHYSIOLOGICAL FACTORS...

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- ❑ **Propulsive movements** will determine *intestinal transit rate* and therefore the *residence time* of a drug in the intestine.
- ❑ This residence time is important since it will dictate the amount of time the dosage form has in which to release the drug, permit dissolution, and allow for absorption.
- ❑ Obviously, the **greater the intestinal motility**, the **shorter the residence time** and the **less time** there is for those processes to proceed.

PHYSIOLOGICAL FACTORS...

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- ❑ **Mixing movements** of the small intestine are a result of contractions dividing a given region of the intestine into segments.
- ❑ These contractions
 - result in the mixing of the intestinal contents with secretions several times a minute.
 - bring the gut contents into optimal contact with the surface epithelium and thereby provide a larger effective area for absorption.

PHYSIOLOGICAL FACTORS...

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- ❑ Transit through the small intestine appears to be quite different in a variety of ways from movement through the stomach.
 - ❖ Small intestinal transit time is relatively constant
- ❑ While this value may range from about 1 to 6 hours, intestinal residence time appears to be relatively consistent among normal subjects.

PHYSIOLOGICAL FACTORS...

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- ❑ Transit appears to be less dependent upon physical nature of the material (liquid vs. solid and size of solids) compared to the response of the stomach.
- ❑ Food appears not to influence intestinal transit as it does gastric emptying.
- ❑ SI does not discriminate b/n DFs, or b/n fed and fasted states

PHYSIOLOGICAL FACTORS...

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❑ ***SITT*** is particularly important for

- Dosage forms that release drug slowly (CR, SR DFs)
- Require time to initiate release (EC DFs, release drug in SI);
- Drugs that dissolve slowly in intestinal fluids; and
- Drugs that are maximally absorbed in certain regions of intestine (carrier-mediated transporters)

PHYSIOLOGICAL FACTORS...

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Colonic Transit

- ❑ Although there are mixing and propulsive movements in the colon, they tend to be rather sluggish.
- ❑ These movements increase the surface area of the colon and result in efficient water absorption.
- ❑ The greatest proportion of time moving down the GIT is spent by a meal moving through the colon.
- ❑ Colonic residence time is considerably longer than in other parts of the GIT, and it is also more variable.
- ❑ *Mouth-to-anus transit time is normally longer than 24 hrs*

PHYSIOLOGICAL FACTORS...

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Food

- ❑ Food influence rate and extent of absorption (direct or indirect)

** Complexation of drugs with components in diet*

- ❑ Becomes issue when irreversible or insoluble complex is formed
- ❑ The absorption of tetracycline is reduced by calcium salts present in dairy foods and by several other cations, including magnesium and aluminium, which are present in antacid preparations.
- ❑ These cations react with tetracycline to form a ***water-insoluble*** and ***non-absorbable complex***.

PHYSIOLOGICAL FACTORS...

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** Alteration of pH*

❑ Food increase stomach pH

→ Decrease rate of dissolution and absorption of **weakly basic** drugs and increase that of **weakly acidic** ones

** Alteration of gastric emptying*

- ✓ Food may reduce the rate or extent of absorption by virtue of reduced gastric emptying rate, which is particularly important for compounds unstable in gastric fluids
- ✓ Fatty foods reduce gastric emptying and delay onset of action

PHYSIOLOGICAL FACTORS...

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* *Stimulation of GI secretions*

- Pepsins produced in response to food may degrade drugs
- Fats stimulate secretion of bile
 - Increase dissolution of poorly soluble drugs and enhance their absorption
 - Bile salts form insoluble and non-absorbable complexes with some drugs, such as *neomycin and nystatin*

PHYSIOLOGICAL FACTORS...

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**Competition b/n food and drugs for specialized absorption mechanisms*

- ❑ Competitive inhibition of absorption of drugs with chemical structure similar to nutrients
- ❑ **L-Dopa** absorption may be inhibited in the presence of certain **amino acids** formed from the digestion of proteins.

**Increased viscosity of GI contents*

- ❑ Food provides viscous environment.
 - reduce rate of drug dissolution
 - diffusion from lumen to absorbing membrane lining GIT
 - ➡ Decrease bioavailability!

PHYSIOLOGICAL FACTORS...

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** Food-induced changes in blood flow*

- ❑ Blood flow to GIT and liver increases after meal and may last several hours
 - Increase rate at which drug is presented to liver
 - Larger fraction of drug escapes first-pass metabolism
- Propranolol

PHYSIOLOGICAL FACTORS...

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* *Food-induced changes in presystemic metabolism*

- ❑ Grapefruit juice inhibits intestinal cytochrome P450 (CYP3A family)

→ ↑ bioavailability of susceptible drugs

*artemether, saquinavir
felodipine, diazepam.
cisapride, cyclosporine*



★ *As a general rule, drugs are better absorbed under fasting conditions.*

PHYSIOLOGICAL FACTORS...

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The unstirred water layer

- It is a more or less stagnant layer of water, mucus and glycocalyx (glycoprotein) adjacent to intestinal wall.
- This layer can provide a diffusion barrier to drugs.
- Some drugs (e.g., quaternary ammonium compounds) are capable of complexing with mucin, thereby reducing their availability for absorption.

PHYSIOLOGICAL FACTORS...

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Disease state

- ❑ Several disease states can influence the rate and extent of drug absorption
- ❑ The 3 major classes of disease that can influence the BA of drugs are:

1. GI diseases

2. CV diseases

3. Hepatic diseases

PHYSIOLOGICAL FACTORS...

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Presystemic metabolism

Def: The metabolism of orally administered drugs by gastrointestinal and hepatic enzymes, resulting in a significant reduction of the amount drug reaching the systemic circulation.

** Gut wall metabolism*

- Mucosal cells lining the gut wall represent a significant potential site for drug metabolism.
- Many drugs have been shown to undergo metabolism in the gut wall, including those that are substrates for the *CYP3A4* isozyme.

e.g. *aspirin, acetaminophen, morphine.*

PHYSIOLOGICAL FACTORS...

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* *Hepatic metabolism*

- After a drug is swallowed, it is absorbed by the digestive system and enters the *hepatic portal system*. It is carried through the portal vein into the liver before it reaches the rest of the body.
- This *first pass* through the liver thus greatly reduces the BA of the drug.
- drugs susceptible to large first-pass effect:
Propranolol, lidocaine, imipramine

